

SULIT
SM015
Mathematics 1
Semester 1
Session 2022/2023

SM015
Matematik 1
Semester 1
Sesi 2022/2023



KEMENTERIAN PENDIDIKAN MALAYSIA

KOLEJ MATRIKULASI KEJURUTERAAN KEDAH

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION
PRA PSPM 1

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

Kertas soalan ini mengandungi **5** halaman bercetak.
This question paper consists 5 printed pages.

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **Section A** and **Section B**.

Section A consists of 3 questions. **Section B** consists of 7 questions.
Answer all questions.

Marks for each question or section are shown in the bracket at the end of the question or section

All steps must be shown clearly.

Only non-programmable scientific calculator can be used.

Numerical answer may be given in the form of π , e , surd, simplest fractions or up to three significant figures, where appropriate, unless stated otherwise in the question.

ARAHAN KEPADA CALON

Kertas soalan ini mengandungi **Bahagian A** dan **Bahagian B**
Bahagian A mengandungi 3 soalan. **Bahagian B** mengandungi 7 soalan.

Jawab semua soalan.

Markah yang diperuntukan bagi setiap soalan atau bahagian soalan ditunjukkan dalam kurungan pada penghujung soalan atau bahagian soalan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas.

Kalkulator saintifik yang tidak boleh diprogramkan sahaja yang boleh digunakan.

Jawapan berangka boleh diberi dalam bentuk π , e , surd, pecahna termudah atau sehingga tiga angka bererti, dimana-mana yang sesuai, kecuali jika dinyatakan dalam soalan.

LIST OF MATHEMATICAL FORMULAE

Quadratic equation $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Differentiation

$f(x)$	$f'(x)$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

Sphere	$V = \frac{4}{3} \pi r^3$	$S = 4\pi r^2$
Right circular cone	$V = \frac{1}{3} \pi r^2 h$	$S = \pi r^2 + \pi r l$
Right circular cylinder	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1 (a) Given

$$f(x) = \begin{cases} ax^2 + 2 & x \leq 2 \\ -3x + 10 & 2 < x < 3 \\ -x + 2 & 3 \leq x \end{cases}$$

is continuous at $x = 2$. Find the value of a . Hence, determine whether f is continuous at $x = 3$.

[6 marks]

(b) Evaluate $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$

[3 marks]

2 (a) Differentiate $y = x^2 + 2x$ by using first principle.

[4 marks]

(b) Find $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$ for $y = \frac{2}{\sqrt{x}} - \sqrt{x}$

[3marks]

(c) Differentiate $\cos(x + y) + 2x \sin y$ with respect x .

[4 marks]

3 Given that $f(x) = \cos 2x, 0 < x < \pi$. Find the relative extrema by using second derivative test.

[5 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

- 1 Given that $z = x + yi$ is a non-zero complex number.
Find the value of x and y such that $z(3+i) = 5(\bar{z} - 2i)(1-i)$. [6 marks]
Hence find the argument of z . [3 marks]

- 2 a) Solve the equation $e^x - 12e^{-x} - 1 = 0$. [4 marks]
b) Find the range of values of x which satisfy $3 < x^2 - 2x < 12 + 2x$. [11 marks]

- 3 The function f is defined as $f : x \rightarrow x^2 - 4x + 5, x \in \mathbb{R}$.
a) Determine whether the inverse of the function f exists or not.
Give reasons for your answer. [4 marks]
b) If the domain of f is modified to $\{x : x \in \mathbb{R}, x \geq M\}$ so that the inverse function exists, find the smallest values of M . [3 marks]

- 4 a) The function f and g are given by

$$f(x) = e^{2x} + 1 \quad \text{and} \quad g(x) = \frac{1}{2} \ln(x-1).$$
 - (i) Show that $f(x)$ is a one-to-one function. [2 marks]
 - (ii) Determine the inverse function of $g(x)$.
Hence, state the relation between $f(x)$ and $g(x)$. [3 marks]
 - (iii) Find the function $h(x)$ if $(f \circ h)(x) = e^{4x^2} + 1$. [2 marks]

- b) Given that $f(x) = 3\sin x$ and $g(x) = x - \pi$.
 - (i) Find $(f \circ g)(x)$. [2 marks]
 - (ii) Sketch the graph of $(f \circ g)(x)$ for $0 \leq x \leq 2\pi$. [3 marks]
 - (iii) State the range of $(f \circ g)(x)$. [1 mark]

5 a) Evaluate the following limits:

i) $\lim_{x \rightarrow +\infty} \frac{\sqrt{x} - 6}{x + 3}$ [3 marks]

ii) $\lim_{x \rightarrow 4^-} \frac{x^2 - 6x + 8}{|x - 4|}$ [3 marks]

b) Function g is defined as follows:

$$g(x) = \begin{cases} 2x & x \leq 2 \\ mx + n & 2 < x < 5 \\ 3 - x & x \geq 5 \end{cases}$$

Determine the values of constants m and n such that g is continuous

on the interval $(-\infty, \infty)$ [5 marks]

6 a) A function f is defined by $f(x) = |x - 1| + 2$. Is f differentiable in its domain? (5 marks)

b) If $y^2 = 2 \ln(\sqrt{x}y^3)$ for $x > 0$ and $y > 0$. Find $\frac{dy}{dx}$. (4 marks)

c) The parametric equations of a curve are given by $x = \frac{t^2}{4t + 1}$ and $y = 4 - t^2$, where t is non-zero parameter.

i) Find $\frac{dy}{dx}$ in terms of t . (5 marks)

ii) Hence, determine the value of t when $\frac{dy}{dx} = -1$. (2 marks)

7 A water trough of length 1 m has a triangular cross-section in the shape of an equilateral triangle of side 60 cm. Water is filled to the trough at a rate of $50\sqrt{3} \text{ cm}^3 \text{ s}^{-1}$. Find the rate of increase of the depth of the water in the trough when it has been filled for 10 minutes.

[8 marks]